

DEPARTMENT OF EDUCATION/EDUCATION TRAINING

MID SEMESTER EXAMINATION

M.Ed. I Semester | 2026 - 27 (Odd Semester)

PAPER IV - RESEARCH METHODS IN EDUCATION (GENERAL PERSPECTIVES) (MED104)

Max Time: 1 Hour 30 Minutes Max Marks: 30 Marks

Note: All Questions are Compulsory / ~~എല്ലാ ചോദ്യങ്ങളും നിർബന്ധപൂർവ്വമാണ്.~~

Section A -- 9 × 1 = 9 Marks

Attempt all Questions / ~~എല്ലാ ചോദ്യങ്ങളും മറുപടി നൽകുക.~~ - (9 × 1 = 9)

1.a Read the following statements carefully and identify the set of statements that is CORRECT.

1. Philosophy provides the conceptual coordinates for understanding reality, knowledge, values, reasoning and human existence.
2. Metaphysics primarily examines how researchers can determine whether their findings are statistically significant.
3. Epistemology deals with questions concerning how we know what we know and what constitutes valid research.
4. Axiology is concerned with values, while ethics guides moral conduct in research.
5. Positivism assumes that reality is socially constructed and its research goal is to understand deeper meanings.
6. Interpretivism views reality as socially constructed and seeks to understand deeper meaning.
7. Pragmatism is concerned with what practically works and with solving real-world problems.
8. Critical Theory accepts existing power structures as neutral and focuses primarily on discovering universal laws.
9. Logical alignment among Reality, Epistemology/Paradigm and Methodology/Process is important for research.
10. The research process moves from Problem → Questions → Design → Data Collection → Analysis → Interpretation → Knowledge Creation.

- A) 1, 2, 3, 4, 7, 9
B) 1, 3, 4, 6, 7, 9, 10
C) 2, 3, 5, 6, 8, 10
D) 1, 4, 5, 7, 8, 9, 10

1.b Read the following statements carefully and identify the set of statements that is CORRECT.

1. Practitioner research can help teachers move from anecdotal assumptions toward evidence-based practice.
2. Inquiry and curiosity involve identifying a knowledge gap that can be investigated systematically.
3. Weekly low-stakes quizzes are compared with regular homework to examine their effect on final examination performance.
4. A quasi-experimental design requires that every participant must be randomly assigned to the intervention and control groups.
5. Pre-test and final-exam measurements can provide baseline and outcome information for examining change.
6. Statistical significance alone necessarily proves that an intervention is the only cause of the observed improvement.
7. A confounding variable may provide an alternative explanation for an observed relationship between an intervention and an outcome.

8. If increased study hours may influence final examination scores, measuring study time and controlling for it can strengthen the analysis.
9. Research concludes permanently once statistically significant results have been obtained, so further questions are unnecessary.
10. Replication is strengthened when research methods are simple, clearly documented and repeatable by other practitioners.

- A) 1, 2, 3, 5, 7, 8, 10
- B) 1, 3, 4, 6, 7, 9, 10
- C) 2, 3, 5, 6, 8, 10
- D) 1, 4, 5, 7, 8, 9, 10

1.c Read the following research situations carefully and identify the set of situations that correctly represents epistemological concerns.

1. A researcher asks students whether their knowledge of a concept comes mainly from classroom observation and personal experience.
2. A researcher examines whether logical reasoning can provide a valid basis for accepting a conclusion about students' learning.
3. A researcher investigates how students construct their understanding of a difficult concept through their previous experiences and interactions.
4. A researcher studies how social and cultural conditions influence what is accepted as knowledge in a classroom.
5. A researcher asks whether a belief held by students should automatically be treated as true simply because they strongly believe it.
6. A researcher explores whether observation, inference, intuition and logic can be considered different ways through which knowledge is acquired.
7. A researcher examines the relationship between the knower and the phenomenon being studied.
8. A researcher decides that the only purpose of epistemology is to calculate the statistical significance of research findings.
9. A researcher questions what can be known about an educational phenomenon and whether some aspects may remain beyond human knowledge.
10. A researcher examines how to distinguish justified or valid knowledge from false or unsupported beliefs.

- A) 1, 2, 3, 5, 7, 8, 10
- B) 1, 3, 4, 5, 6, 8, 9
- C) 2, 3, 4, 6, 7, 9, 10
- D) 1, 2, 3, 5, 6, 7, 8

1.d Read the story carefully and identify the correct combination of statements.
(Ananya's Case Study on Knowledge Acquisition)

1. Her teacher explains a difficult concept, and Ananya accepts the explanation because she considers the teacher knowledgeable and trustworthy.
2. During a festival, Ananya follows a family ritual that has been practiced across generations.
3. Ananya learns to ride a bicycle through repeated attempts, mistakes and practice.
4. After observing the sunrise repeatedly in the east, she forms a general conclusion about where the sun rises.
5. She reasons: "All humans need food. I am a human. Therefore, I need food."
6. While studying plants, Ananya notices that they grow toward the window, proposes a hypothesis, changes their position and observes the results systematically.
7. Ananya decides that knowledge gained from personal experience is always true and never needs further justification.
8. She realizes that knowledge is not merely information but understanding with meaning.
9. She recognizes that knowledge should be reliable, justified, systematic and

capable of being verified.

10. She concludes that scientific method involves systematic observation, hypothesis, experimentation, analysis and conclusion.

A) 2, 3, 4, 5, 7, 8, 9, 10
B) 1, 2, 3, 4, 6, 7, 8, 10
C) 1, 3, 5, 6, 7, 8, 9
D) 1, 2, 3, 4, 5, 6, 8, 9, 10

1.e Three M.Ed. studentsâ Kunal, Manav and Vivekâ were discussing the meaning of philosophy while preparing for their Research in Education class.

1. Kunal said, "I have read many books and collected a large amount of information. Therefore, I have knowledge."
2. Manav said, "I can solve difficult problems quickly by applying logic, so I am intelligent."
3. Vivek replied, "Knowledge gives us facts, intelligence helps us use them, but wisdom helps us decide what is right and best."
4. The teacher explained that philosophy comes from the Greek words Philos (love) and Sophia (wisdom), meaning "love for wisdom."
5. She further explained that philosophy is not merely collecting facts or becoming intelligent; it involves curiosity, reflection, questioning and the search for truth, goodness and a meaningful life.
6. Kunal argued that once a person possesses sufficient information, there is no need for reflection or questioning.
7. Manav observed that intelligence and wisdom are exactly the same because both involve the ability to solve problems.
8. Vivek used his knowledge and intelligence to make responsible decisions and help others, showing how wisdom can transform knowledge into meaningful action.
9. The teacher concluded that philosophy encourages us to use knowledge and intelligence for a meaningful and better life.
10. She added that philosophical thinking can be expressed as: Philosophy â Search for Wisdom â Better Life â Better World.

A) 2, 3, 4, 5, 6, 7, 9
B) 1, 2, 4, 6, 7, 8, 10
C) 1, 2, 3, 4, 5, 8, 9, 10
D) 1, 3, 5, 6, 7, 8, 9, 10

1.f Aarav, a postgraduate student, was preparing a seminar on how knowledge, education and research contribute to individual and social development.

1. Knowledge represents awareness and understanding of reality and may develop through experience, thinking, reflection and discovery.
2. Education mainly transmits existing knowledge and has no significant role in developing skills, attitudes, values or critical thinking.
3. Research explores unknowns, generates new knowledge and can validate or refine existing knowledge.
4. Education can inspire questions and prepare learnersâ minds for research.
5. Research and education are completely independent processes because education only uses old knowledge whereas research only produces unrelated new knowledge.
6. Knowledge can be organized and transmitted through education, while research can expand and refine that knowledge.
7. The cycle may be understood as Knowledge â Education â Research â New Knowledge â Benefits to Society.
8. Research is useful only for producing academic publications and has no role in innovation, problem-solving or societal development.
9. Knowledge, education and research together can contribute to personal growth, social progress, economic development and a sustainable future.

A) 1, 2, 3, 4, 5, 7, 9
B) 1, 3, 4, 6, 7, 9, 10
C) 2, 3, 5, 6, 8, 9, 10
D) 1, 3, 5, 6, 7, 8, 10

[illegible]

E. Generalization / $\exists x \exists y (A(x) \wedge B(y) \rightarrow C(x, y))$

2. Observations and descriptions should be exact and free from unnecessary ambiguity. /
3. A scientific claim should be open to being tested and potentially shown to be false. /
4. Prefer the simplest adequate explanation when competing explanations are available. /
5. Findings obtained under appropriate conditions may be extended beyond the

- A) Â 1, B 3, C 2, D 4, E 5
B) Â 3, B 1, C 4, D 2, E 5
C) Â 1, B 4, C 2, D 5, E 3
D) Â 5, B 3, C 1, D 4, E 2

A) Both A and R are true, and R is the correct explanation of A.
B) Both A and R are true, but R is not the correct explanation of A.
C) A is true, but R is false.
D) A is false, but R is true.

1. Research generates new ideas.
2. Ideas are important for the growth of knowledge.
3. Research involves synthesis and re-synthesis of knowledge.
4. Research critically examines existing knowledge.
5. Universities can act as knowledge banks.
6. Research helps transmit knowledge from one generation to another.
7. Research investigates existing situations and real-life problems.
8. Scientific research aims at objectivity, reliability, validity and verifiability.
9. Research can provide practical and effective solutions to problems.
10. Research is concerned only with storing existing knowledge and does not explore new solutions.

A) 1, 2, 4, 5, 6, 8, 9, 10
B) 1, 2, 3, 4, 5, 7, 8, 9
C) 2, 3, 5, 6, 7, 8, 10
D) 1, 3, 4, 6, 7, 9, 10

Section B -- 3 x 3 = 9 Marks

Attempt any three of the following Questions / $\frac{3}{3} \times 3 = 9$

2.a A researcher is interested in three different educational situations: A. Understanding how students develop the concept of creativity without an immediate practical application. B. Finding whether a new teaching strategy can improve mathematics achievement in secondary school students. C. Solving a specific classroom problem of poor reading comprehension through a teacher's own intervention. Identify the type of research represented by A, B and C and frame one appropriate research problem for each.

2.b A researcher wants to study "Why do some rural students hesitate to participate in online classes?" He plans to conduct interviews and observe students's experiences rather than collect numerical scores. Apply your understanding of research approaches and identify the most appropriate approach. Give two reasons for your choice.

2.c A researcher finds that students taught through a new teaching strategy perform better in a test. However, instead of claiming that the strategy certainly and completely caused the improvement, the researcher considers possible limitations, alternative explanations and the fallibility of measurement before drawing a conclusion. â Which research approach is reflected in this situation? Explain how the researcherâ s thinking reflects the key assumptions of this approach.

2.d A researcher studies how children develop the concept of "fairness" during different stages of cognitive development, without intending to develop an immediate classroom intervention. What type of research is this? Explain the conceptual basis of your answer and state how the findings may contribute to educational knowledge.

Section C -- 2 ^ 6 = 12 Marks

Attempt any two of the following Questions / $\frac{2}{3} \times 6 = 4$ - (2⁶ = 12)

3.a Imagine that you are investigating education through three different windows: 1. Past: You want to understand how education policies during the British period influenced the development of education in India. 2. Present: You want to find out the present status of digital learning among secondary school students. 3. Intervention: You want to test whether a new teaching method produces better science achievement than the existing method. Identify the research design appropriate for each window and formulate one suitable research problem for each.

3.b A researcher wants to understand how Positivism, Pre-Positivism and Post-Positivism differ in their views about knowledge and research. â Differentiate among the three approaches on the basis of their view of reality, nature of knowledge, role of the researcher, and approach to objectivity. Present your answer in a concise comparative form.

3.c Consider the following research problem: "A Study of the Effect of Weekly Low-Stakes Quizzes on the Final Examination Performance of Class IX Students in Science Compared with Regular Classroom Assessment." â Based on the above research problem, identify and briefly explain its major components: 1.

Population â Who is being studied? 2. Independent Variable â What is being introduced or changed? 3. Dependent Variable â What outcome is being measured? 4. What result is measured? 5. Comparison â What is the intervention being compared with? 6. Context/Subject â In what educational context is the study conducted? 7. Relationship/Effect â What relationship or effect is being investigated?